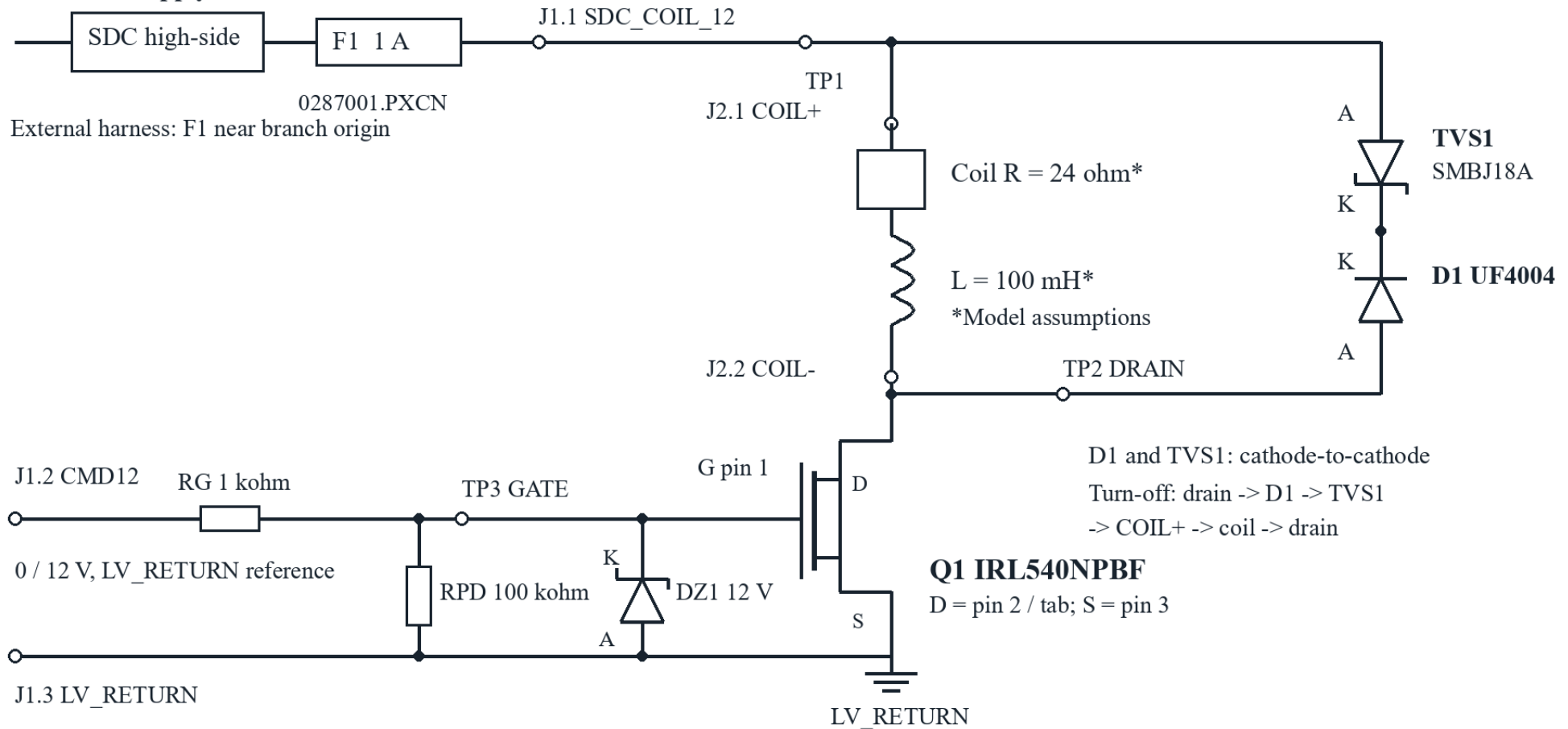


C.1 12 V coil: low-side drive and local turn-off clamp

Schematic proposal | filled dots = connected nets | A = anode; K = cathode

Protected LV supply

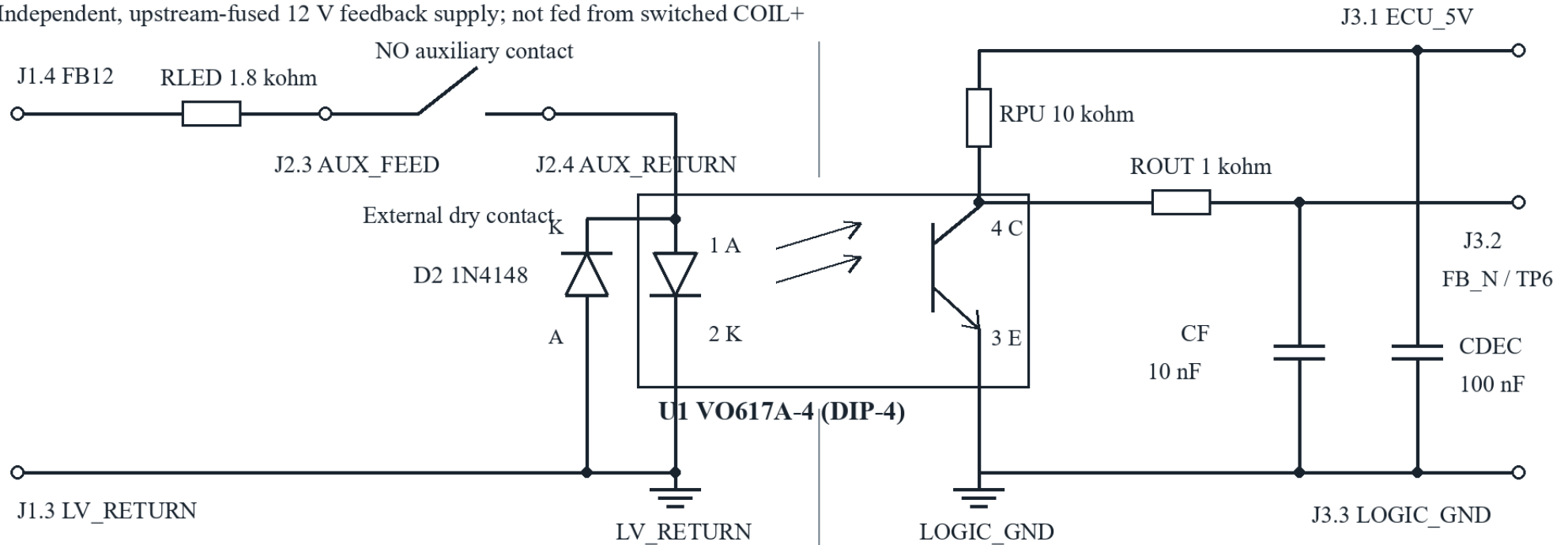


Mount D1/TVS1 close to coil switching loop. Keep gate return separate from coil-current routing.

SDC removes coil power independently of CMD12. This drawing contains no tractive-system wiring.

C.2 Protected auxiliary-contact feedback to 5 V logic

Independent, upstream-fused 12 V feedback supply; not fed from switched COIL+



Optical barrier: no copper / ground bridge on this board

AUX closed: LED on -> transistor on -> FB_N LOW. AUX open / field wire open: FB_N HIGH.

RLED is placed before the external contact to limit current in both field wires. D2 protects LED reversal.

True end-to-end galvanic isolation requires the two grounds to remain unbonded elsewhere.

J1.5 unused. Connector cavity numbering is defined in the pinout table, not by drawing left-to-right order.